

Paramveer Singh Bhele

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Software Engineering | AI Systems | Infrastructure

EDUCATION

Columbia University

August 2026 to December 2027

Master of Science in Artificial Intelligence, Concentration in AI Infrastructure

New York, NY

University of South Florida

August 2022 to May 2026

Bachelor of Science in Computer Science, Minor in Entrepreneurship, GPA 3.71/4.00

Tampa, FL

Coursework: Data Structures & Algorithms, Algorithm Analysis, Software Engineering, OOP in C++, Secure Coding

Recognition: \$43,000 in merit scholarships, AI4ALL Ignite Fellow, MLH Best Use of Gemini API winner

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

AI / ML: PyTorch, TensorFlow, Scikit-learn, Transformers, BERT, LoRA, LLM APIs, embeddings, model evaluation

Backend / Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MongoDB, SQL, data pipelines

Product / Infrastructure: React, Next.js, Git, GitHub Actions, Docker, AWS, Google Cloud, Vercel, Render, CI/CD

EXPERIENCE

Student Assistant - Digital Initiatives

August 2024 to May 2026

University of South Florida Libraries

Tampa, FL

- Cut manual metadata review time by approximately 70% by building scraping and validation tooling for digital collections
- Improved data quality across 1,000+ digital records by standardizing metadata and resolving validation inconsistencies

Machine Learning Engineer Intern

October 2025 to January 2026

Finds, Inc.

Remote

- Improved image-processing workflows by integrating in-house ML models into Python services backed by PostgreSQL
- Increased data consistency by designing schemas, metadata pipelines, scrapers, and automated validation checks
- Accelerated production iteration by converting model outputs into structured records for downstream application workflows

Undergraduate Research Assistant

February 2025 to July 2025

Reality, Autonomy, and Robot Experience Lab, University of South Florida

Tampa, FL

- Built an Android research app integrating Gemini and Google Cloud to support structured AI-assisted study workflows
- Improved reproducibility across HRI studies by collecting, validating, and analyzing experimental data and procedures

App Developer Intern

May 2022 to July 2022

Coefficient Software Systems Pvt. Ltd.

Navi Mumbai, India

- Improved usability of a 100,000+ download consumer app by refining interface flows and supporting production updates
- Expanded benchmarking coverage to 500+ universities across three markets by structuring and validating product data

PROJECTS

Polaris | *Python, FastAPI, Next.js, PostgreSQL, GitHub App, Gemini* | [Live](#) | [GitHub](#)

- Cut IaC pull-request security review to under 60 seconds by combining deterministic checks with dual Gemini agents
- Improved fix trust by separating patch generation and verification with repository isolation and developer approval
- Shipped control mapping, PostgreSQL findings, PR comments, audit trails, and verified fixes across four IaC formats

PR Nutrition | *TypeScript, Node.js, GitHub Actions, Vitest* | [GitHub](#) | [npm](#)

- Reduced PR review overhead by building a deterministic CLI and GitHub Action that prioritizes higher-risk changes
- Shipped v0.4.0 to npm with local-first analysis, Markdown/JSON reports, CI integration, and zero external model calls

Adversarial Spam Detection | *Python, PyTorch, BERT, Qwen3-4B, LoRA* | [GitHub](#)

- Raised adversarial detection from 91.7% to 95.7% across three rounds by generating attacks and retraining BERT
- Maintained 96.5% clean accuracy while improving robustness through LoRA fine-tuning and controlled evaluation

CarbonCTRL | *React, TypeScript, Express, MongoDB, Gemini, Python ML* | [GitHub](#)

- Won MLH Best Use of Gemini API among 63 submissions by shipping a full-stack emissions platform at HackaBull 2025
- Turned emissions inputs into estimates and reduction plans through React, APIs, MongoDB, and Gemini workflows

LEADERSHIP AND RECOGNITION

Hackathon Judge | *HealthHacks 2025*

October 2025

- Evaluated healthcare technology projects by assessing technical execution, user impact, usability, and product viability

Head of Logistics | *TEDx at USF*

June 2024 to January 2025

- Coordinated 10+ volunteers and venue operations to deliver a 100-seat TEDx event across multiple teams

Vice President | *Students of India Association at USF*

January 2024 to May 2024

- Raised \$15,000 and delivered events for 1,000+ attendees by leading a 26-member board and partner outreach